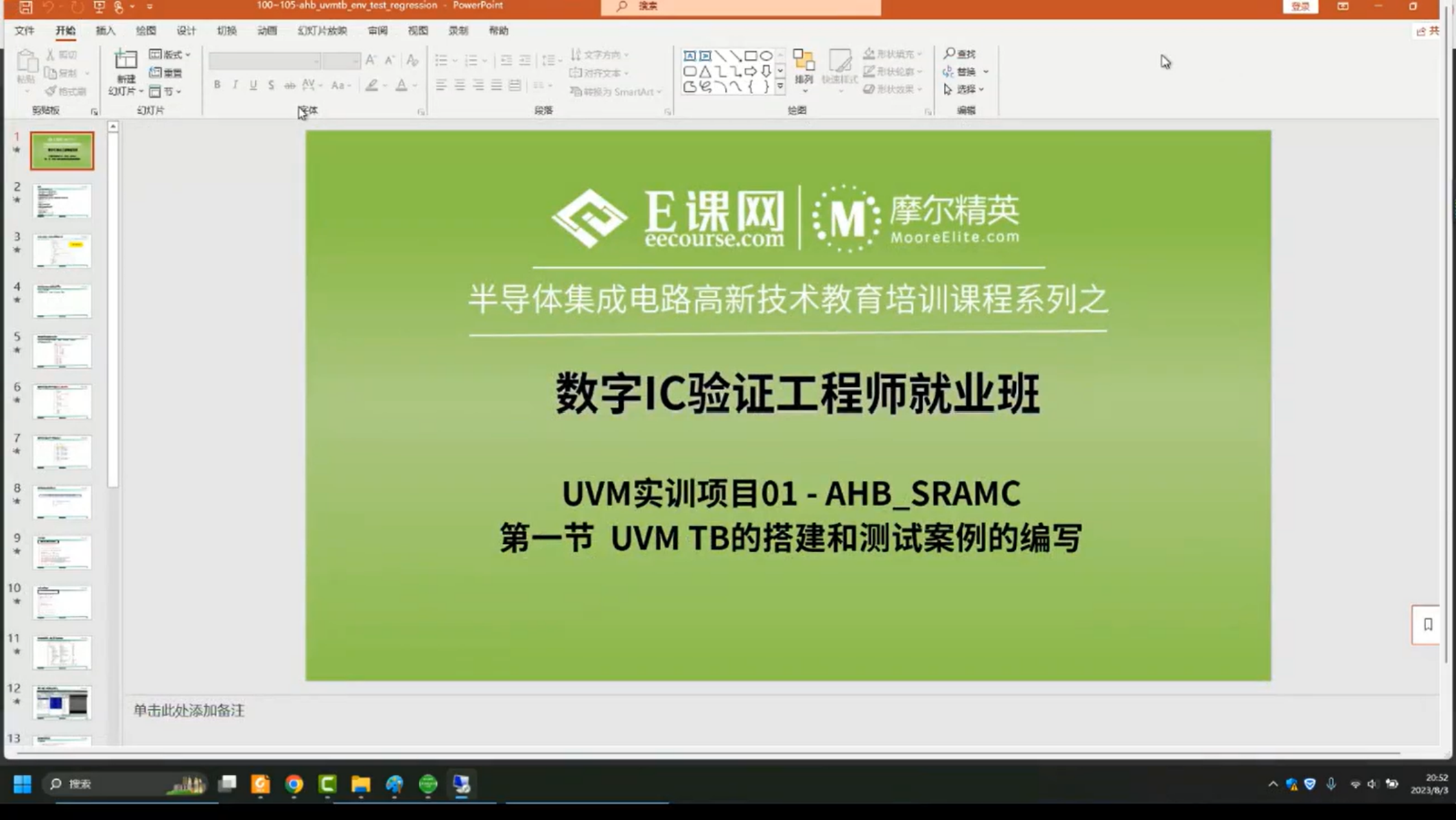

100-105-ahb_uvmtb_env_test_regression - PowerPoint
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UVM实训项目01 - AHB_SRAMC
第一节 UVM TB的搭建和测试案例的编写
单击此处添加备注

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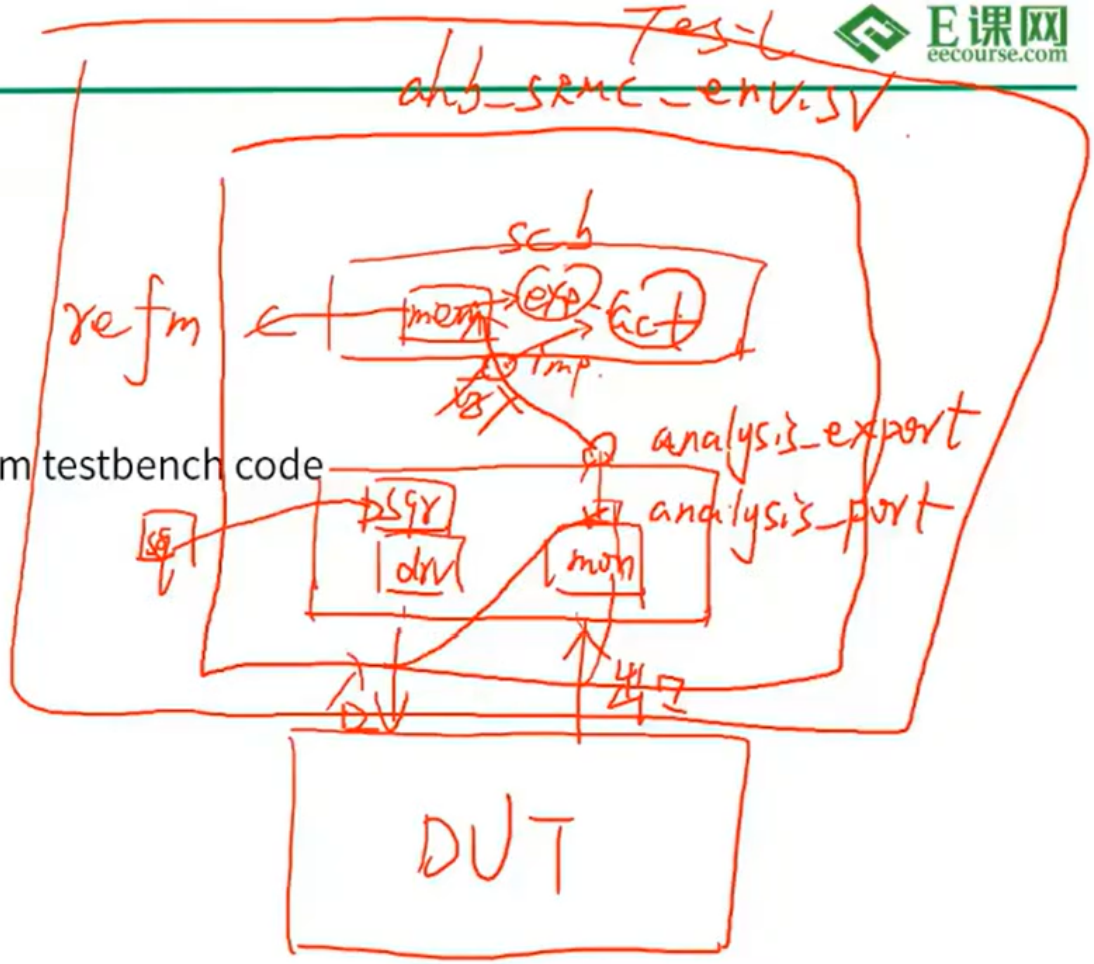
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UVM实训项目01 - AHB_SRAMC

第一节 UVM TB的搭建和测试案例的编写

内容

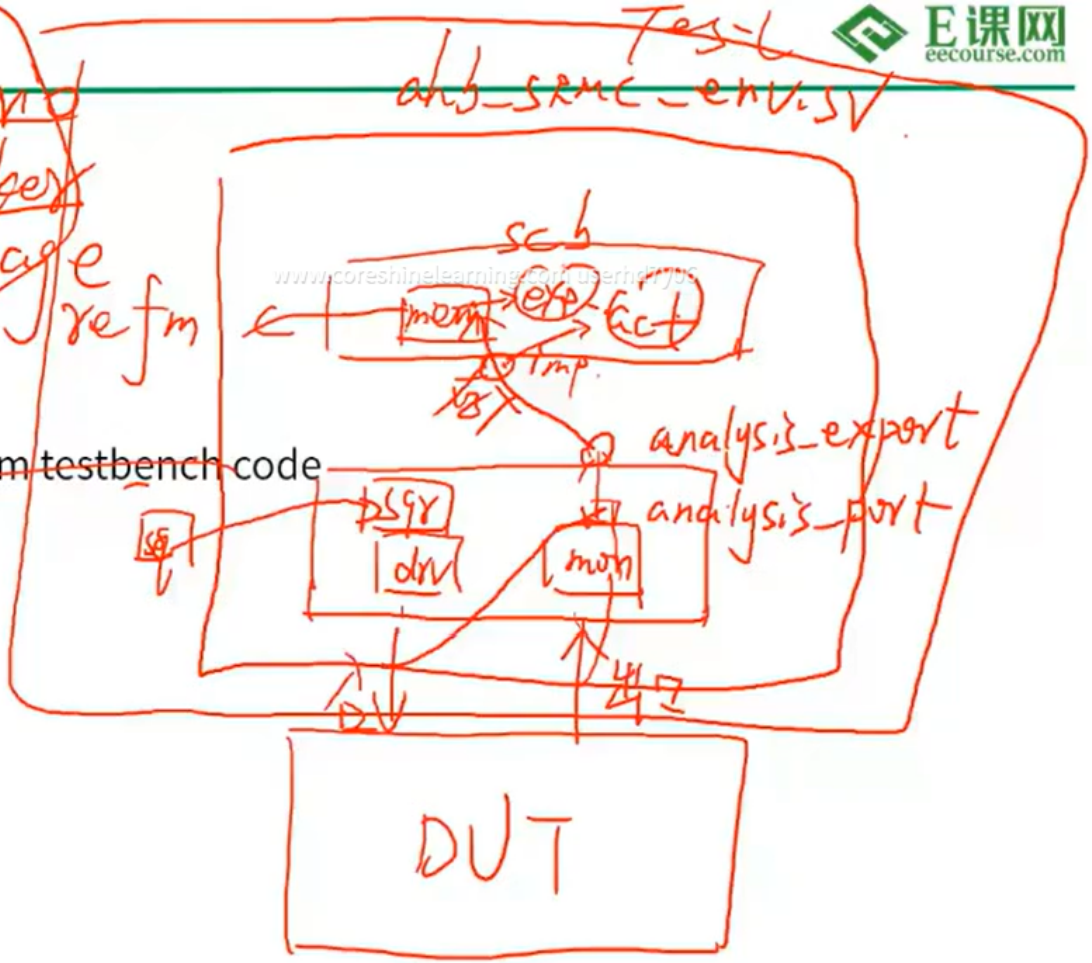
- AHB-SRAMC UVMTB验证计划
- AHB-SRAMC验证平台UVMTB的架构图
- AHB-SRAMC验证平台UVMTB的代码目录
- 如何构建通用的UVM Testbench
- 利用perl脚本 uvm_tb_gen.pl生产通用型的uvm testbench code
- 如何编写interface code?
- 如何编写driver?
- 如何编写sequence item?
- 如何编写sequence & testcase?
- 如何为激励添加constraints?
- 如何建立regression?
- 如何统计和分析code coverage?
- 如何查看testcase passing rate?



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内容

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4 UVM testbench 验证平台 .....
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4.2 验证方法： UVM .....
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详细介绍验证计划

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如何构建UVMTB的通用型代码

E课网 eecourse.com

- uvm_tb_gen.pl
- 讲解使用方法： ./uvm_tb_gen.pl -help

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如何编写interface code?

E课网 eecourse.com

- 参考设计文档中输入和输出： logic、clocking、modport
- DUT的顶层信号接口

```
20 `ifndef AHB_IF_SV
21 `define AHB_IF_SV
22
23 interface ahb_if(input hclk,input hresetn);
24   logic   hsel;
25   logic   hwrite;
26   logic   hready;
27   logic [2:0]   hsize;
28   logic [1:0]   htrans;
29   logic [2:0]   hburst;
30   logic [31:0]  hwdata;
31   logic [31:0]  haddr;
32   logic   hready_resp;
33   logic [1:0]   hresp;
34   logic [31:0]  hrdata;
35
36 clocking drvCLK @(posedge hclk);
37   output  hsel;
38   output  hwrite;
39   output  hready;
40   output  hsize;
41   output  htrans;
42   output  haddr;
43   output  hwdata;
44   input   hresetn;
45 endclocking
```

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如何在tb top文件中生成clock & reset?

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```
33  bit hclk;
34  bit sram_clk;
35  bit hresetn;
36
37  ahb_if    m_ahb_if(hclk,hresetn);
38
39  //HCLK 50M
40  initial begin
41    hclk = 0;
42    forever begin
43      #10 hclk = ~hclk;
44    end
45  end
46
47  initial begin
48    sram_clk = 1;
49    forever begin
50      #10 sram_clk = ~sram_clk;
51    end
52  end
53
54  initial begin
55    hresetn = 0;
56    #200;
57    hresetn = 1;
58  end
```


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32:17 / 55:45 护眼: 关 x2倍速 高清

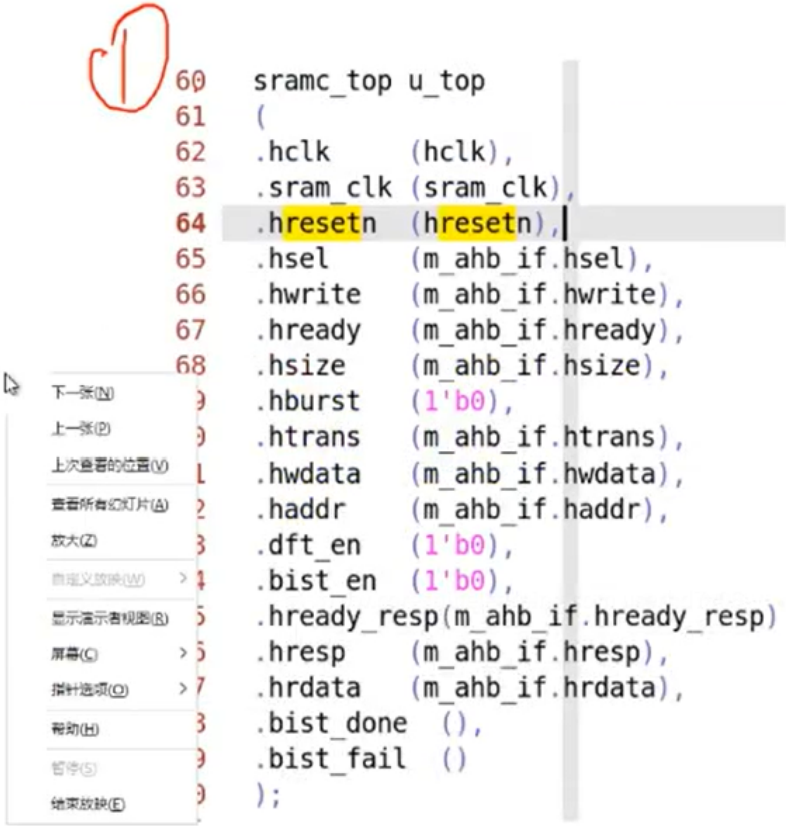
如何在tb top文件中生成clock & reset?




```
33  bit hclk;
34  bit sram_clk;
35  bit hresetn;
36
37  ahb_if    m_ahb_if(hclk,hresetn);
38
39  //HCLK 50M
40  initial begin
41    hclk = 0;
42    forever begin
43      #10 hclk = ~hclk;
44    end
45  end
46
47  initial begin
48    sram_clk = 1;
49    forever begin
50      #10 sram_clk = ~sram_clk;
51    end
52  end
53
54  initial begin
55    hresetn = 0;
56    #200;
57    hresetn = 1;
58  end
```


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如何在tb top文件中例化DUT?



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如何在tb top文件中例化DUT?



```
60  sramc_top u_top
61  (
62  .hclk      (hclk),
63  .sram_clk  (sram_clk),
64  .hresetn   (hresetn),
65  .hsel      (m_ahb_if.hsel),
66  .hwrite    (m_ahb_if.hwrite),
67  .hready    (m_ahb_if.hready),
68  .hsize     (m_ahb_if.hsize),
69  .hburst    (1'b0),
70  .htrans    (m_ahb_if.htrans),
71  .hwdata    (m_ahb_if.hwdata),
72  .haddr     (m_ahb_if.haddr),
73  .dft_en    (1'b0),
74  .bist_en   (1'b0),
75  .hready_resp(m_ahb_if.hready_resp)
76  .hresp     (m_ahb_if.hresp),
77  .hrdata    (m_ahb_if.hrdata),
78  .bist_done  (),
79  .bist_fail  ()
80  );
```



run_test()

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如何dump waveform?

E课网 eecourse.com

dump fsdb vars

vcs -sverilog –ntb_opts uvm –debug_all –timescale=1ns/1ps

```
91   initial begin
92     //if($test$plusargs("VPD_ON"))begin
93         $vcdpluson();
94     //    $vcdplusmemon();
95     //end
96   end
```

Dump fsdb vars

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Test base

uvm_ahb_sram_test_base.sv

```
20 `ifndef UVM_SRAMC_TEST_BASE_SV
21 `define UVM_SRAMC_TEST_BASE_SV
22
23 class uvm_sramc_test_base extends uvm_test;
24
25   `uvm_component_utils(uvm_sramc_test_base)
26
27   uvm_sramc_env           m_env;
28   uvm_sramc_env_config    m_env_config;
29
30   ahb_agent_config  m_ahb_agent_config;
31
32   extern function new(string name, uvm_component parent=null);
33   extern function void build_phase(uvm_phase phase);
34   extern function void connect_phase(uvm_phase phase);
35   extern function void end_of_elaboration_phase(uvm_phase phase);
36   extern task          main_phase(uvm_phase phase);
37
38 endclass : uvm_sramc_test_base
```

base_case

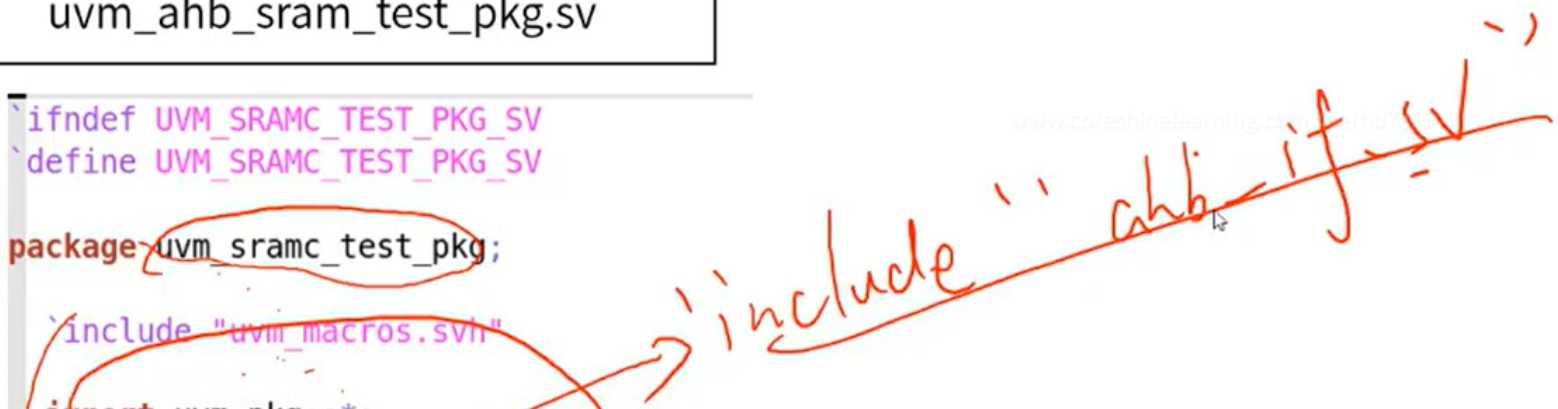
35:52 / 55:45 护眼: 关 x2倍速 高清

Test package

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uvm_ahb_sram_test_pkg.sv

```
20 `ifndef UVM_SRAMC_TEST_PKG_SV
21 `define UVM_SRAMC_TEST_PKG_SV
22
23 package uvm_sramc_test_pkg;
24
25   `include "uvm_macros.svh"
26
27   import uvm_pkg::*;
28
29   import ahb_pkg::*;
30   import uvm_sramc_env_pkg::*;
31
32   `include "ahb_seq_rw.sv"
33
34   `include "uvm_sramc_test_base.sv"
35   `include "sramc_test.sv"
36
37 endpackage : uvm_sramc_test_pkg
38
39 `endif // UVM_SRAMC_TEST_PKG_SV
```



使用Makefile，确认tb topology

E课网 eecourse.com

```
40 UVM_INFO @ 0: reporter [UVMTOP] UVM testbench topology:
41 ----------------------------------------------------------------
42 Name                        Type                      Size  Value
43 ----------------------------------------------------------------
44 uvm_test_top                sramc_test                -     @462
45   m_env                     sramc_1_env               -     @470
46     m_ahb_agent             ahb_agent                 -     @514
47       analysis_port         uvm_analysis_port         -     @522
48       m_driver              ahb_driver                -     @556
49         rsp_port            uvm_analysis_port         -     @573
50         seq_item_port       uvm_seq_item_pull_port    -     @564
51       m_monitor             ahb_monitor               -     @539
52         analysis_port       uvm_analysis_port         -     @547
53       m_sequencer           ahb_sequencer             -     @582
54         rsp_export          uvm_analysis_export       -     @590
55         seq_item_export     uvm_seq_item_pull_imp     -     @696
56         arbitration_queue   array                     0     -
57         lock_queue          array                     0     -
58         num_last_reqs       integral                  32    'd1
59         num_last_rsps       integral                  32    'd1
60       is_active             uvm_active_passive_enum   1     UVM_ACTIVE
61       m_cfg                 ahb_agent_config          -     @538
62     m_refm                  sramc_1_refm              -     @489
63     m_scb                   sramc_1_scb               -     @497
64       analysis_imp          uvm_analysis_imp          -     @505
65 ----------------------------------------------------------------
```


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如何编写driver?

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- 完成reset

```
55 task ahb_driver::reset_phase(uvm_phase phase);
56   phase.raise_objection(this);
57  if(vif.hresetn==0) begin
58       vif.drvCLK.hwrite<=0;
59       vif.drvCLK.hsel<=0;
60       vif.drvCLK.htrans<=0;
61       vif.drvCLK.hready<=0;
62       vif.drvCLK.hsize<=0;
63      // @(vif.drvCLK)
64       vif.drvCLK.haddr<=0;
65     //  @(vif.drvCLK)
66       vif.drvCLK.hwdata<=0;
67     end
68 phase.drop_objection(this);
69 endtask : reset_phase
```


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如何编写driver?

- 从简单开始：2beat timing
- 在考虑连续传输等复杂情况

```
84
85 task ahb_driver::do_drive(ahb_seq_item req);
86  `uvm_info("TRACE",$sformatf("%m"),UVM_HIGH);
87
88       @(vif.drvCLK)
89       vif.drvCLK.hwrite<=req.hwrite;
90       vif.drvCLK.hsel<=1;
91       vif.drvCLK.htrans<=2'b10;
92       vif.drvCLK.hready<=1;
93       vif.drvCLK.hsize<=req.hsize;
94      // @(vif.drvCLK)
95       vif.drvCLK.haddr<=req.haddr;
96       @(vif.drvCLK)
97       vif.drvCLK.hwdata<=req.hwdata;
98
99  `uvm_info("TRACE","driver data",UVM_HIGH);
100
101 endtask : do_drive
```

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如何编写sequence item（transaction/packet）？

- 根据ahb slave的AMBA AHB 2.0的协议
- 根据sramc的详细设计文档

```
23 class ahb_seq_item extends uvm_sequence_item;
24
25 rand bit [15:0] haddr;
26 rand bit [2:0]  hsize;
27 rand bit [1:0]  htrans;
28 rand bit        hsel;
29 rand bit        hready;
30 rand bit        hwrite;
31 rand bit [31:0] hwdata;
32 rand bit [31:0] hrdata;
33 //rand bit        hresetn;
34   `uvm_object_utils_begin(ahb_seq_item)
35  `uvm_field_int(haddr,UVM_ALL_ON)
36       `uvm_field_int(hsize,UVM_ALL_ON)
37       `uvm_field_int(htrans,UVM_ALL_ON)
38       `uvm_field_int(hwrite,UVM_ALL_ON)
39       `uvm_field_int(hwdata,UVM_ALL_ON)
40       `uvm_field_int(hrdata,UVM_ALL_ON)
41   `uvm_object_utils_end
42
43   constraint htr_c  {htrans==2'b10; }
44   constraint hsize_c  { hsize inside {2'b00,2'b01,2'b10}; }
45   extern function new(string name = "ahb_seq_item");
46 endclass : ahb_seq_item
47
48 function ahb_seq_item::new(string name = "ahb_seq_item");
49   super.new(name);
50 endfunction : new
```

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如何约束输入激励?

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- 在sequence中添加约束信息

```
17 task ahb_seq_rw::body();
18   `uvm_info(get_type_name(), "Default sequence starting***", UVM_HIGH)
19
20    for(int i=0;i<20;i++)
21      `uvm_do_with(req,{haddr==i;hwrite==1;hsize==0;});
22
23   for(int i=0;i<20;i++)
24      `uvm_do_with(req,{haddr==i;hwrite==0;hsize==0;});
25
26   `uvm_info(get_type_name(), "Default sequence completed", UVM_HIGH)
27 endtask : body
```

如何编写testcase?

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- 首先确定激励：sequence
- 然后在testcase的不同的phase执行不同的sequence

```
38 task sramc_test::main_phase(uvm_phase phase);
39   ahb_seq_rw seq1;
40   uvm_objection objection;
41   super.main_phase(phase);
42   objection = phase.get_objection();
43   objection.set_drain_time(this,3us);
44   seq1 = ahb_seq_rw::type_id::create("seq1");
45
46   phase.raise_objection(this, "TEST Started!");
47   seq1.start(m_env.m_ahb_agent.m_sequencer);
48   phase.drop_objection(this,"Test Finished!");
49 endtask : main_phase
```

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如何约束输入激励?

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- 在sequence中添加约束信息

```
17 task ahb_seq_rw::body();
18   `uvm_info(get_type_name(), "Default sequence starting***", UVM_HIGH)
19
20    for(int i=0;i<20;i++)
21      `uvm_do_with(req,{haddr==i;hwrite==1;hsize==0;});
22
23   for(int i=0;i<20;i++)
24      `uvm_do_with(req,{haddr==i;hwrite==0;hsize==0;});
25
26   `uvm_info(get_type_name(), "Default sequence completed", UVM_HIGH)
27 endtask : body
```


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如何建立self-checking? scoreboard

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```
57 function void sramc_1_scb::compare(ahb_seq_item tr);
58   int index[$];
59   index=tr_q.find_index() with ((item.haddr == tr.haddr));
60   if((index.size==0)) begin
61   // `uvm_error("transation not found",tr.sprint());
62    return;
63    end
64
65 if(tr_q[index[index.size-1]].hsize < tr.hsize)
66   hsize_comp = tr.hsize;
67 else
68   hsize_comp = tr_q[index[index.size-1]].hsize;
69
70 //if(tr_q[index[index.size-1]].hsize == 2'b10) begin  //check word
71   if(hsize_comp == 2'b10) begin
72     if(tr_q[index[index.size-1]].hwdata==tr.hrdata) begin
73     count_m++;
74     //`uvm_info("TRACE MATCH",{"\nwrite",tr_q[index[index.size-1]].sprint(),"\nread by word",tr.sprint()},UVM_LOW)
75   end
76   else  begin
77     count_dism++;
78     //`uvm_info("TRACE DISMATCH",{"\nwrite",tr_q[index[index.size-1]].sprint(),"\nread by word",tr.sprint()},UVM_LOW)
79   end
80   tr_q.delete(index[index.size-1]);
81 end
82
```

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如何编写testcase?

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- 首先确定激励：sequence
- 然后在testcase的不同的phase执行不同的sequence

```
38 task sramc_test::main_phase(uvm_phase phase);
39   ahb_seq_rw seq1;
40   uvm_objection objection;
41   super.main_phase(phase);
42   objection = phase.get_objection();
43   objection.set_drain_time(this,3us);
44   seq1 = ahb_seq_rw::type_id::create("seq1");
45
46   phase.raise_objection(this, "TEST Started!");
47   seq1.start(m_env.m_ahb_agent.m_sequencer);
48   phase.drop_objection(this,"Test Finished!");
49 endtask : main_phase
```

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如何建立self-checking? scoreboard

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```
57 function void sramc_1_scb::compare(ahb_seq_item tr);
58   int index[$];
59   index=tr_q.find_index() with ((item.haddr == tr.haddr));
60   if((index.size==0)) begin
61   // `uvm_error("transation not found",tr.sprint());
62    return;
63    end
64
65 if(tr_q[index[index.size-1]].hsize < tr.hsize)
66   hsize_comp = tr.hsize;
67 else
68   hsize_comp = tr_q[index[index.size-1]].hsize;
69
70 //if(tr_q[index[index.size-1]].hsize == 2'b10) begin  //check word
71   if(hsize_comp == 2'b10) begin
72     if(tr_q[index[index.size-1]].hwdata==tr.hrdata) begin
73     count_m++;
74     //`uvm_info("TRACE MATCH",{"\nwrite",tr_q[index[index.size-1]].sprint(),"\nread by word",tr.sprint()},UVM_LOW)
75   end
76   else  begin
77     count_dism++;
78     //`uvm_info("TRACE DISMATCH",{"\nwrite",tr_q[index[index.size-1]].sprint(),"\nread by word",tr.sprint()},UVM_LOW)
79   end
80   tr_q.delete(index[index.size-1]);
81 end
82
```


40:53 / 55:45 护眼 关 x2倍速 高清

如何建立self-checking? scoreboard

```
57 function void sramc_1_scb::compare(ahb_seq_item tr);
58   int index[$];
59   index=tr_q.find_index() with ((item.haddr == tr.haddr));
60   if((index.size==0)) begin
61   // `uvm_error("transation not found",tr.sprint());
62    return;
63    end
64
65 if(tr_q[index[index.size-1]].hsize < tr.hsize)
66   hsize_comp = tr.hsize;
67 else
68   hsize_comp = tr_q[index[index.size-1]].hsize;
69
70 //if(tr_q[index[index.size-1]].hsize == 2'b10) begin  //check word
71   if(hsize_comp == 2'b10) begin
72     if(tr_q[index[index.size-1]].hwdata==tr.hrdata) begin
73     count_m++;
74     //`uvm_info("TRACE MATCH",{"\nwrite",tr_q[index[index.size-1]].sprint(),"\nread by word",tr.sprint()},UVM_LOW)
75   end
76   else  begin
77     count_dism++;
78     //`uvm_info("TRACE DISMATCH",{"\nwrite",tr_q[index[index.size-1]].sprint(),"\nread by word",tr.sprint()},UVM_LOW)
79   end
80   tr_q.delete(index[index.size-1]);
81 end
82
```

www.coreshulearning.com usemefy05

如何建立regression?

- 需要多个testcase
- 使用脚本运行regression
- 一般每天运行一次，或者有RTL修改之后，运行一次

www.coreshinelearning.com userhd7y06

49:32 / 55:45 护眼: 关 x2倍速 高清

如何统计和分析RTL代码覆盖率？

E课网 eecourse.com

```
19 COV =-cm line+cond+fsm+tgl -cm_dir ./logs/COV
20 COV +=  -cm_hier exclude_bist.fl


23 regr:
24   make comp
25   make run TEST=sramc_test_addr_100  N=0
26   make run TEST=sramc_test_rand_hsize HSIZE_W=0 HSIZE_R=0 N=1
27   make run TEST=sramc_test_rand_hsize HSIZE_W=1 HSIZE_R=1 N=2
28   make run TEST=sramc_test_rand_hsize HSIZE_W=2 HSIZE_R=2 N=3
29   make run TEST=sramc_test_rand_hsize HSIZE_W=1 HSIZE_R=2 N=4
30   make run TEST=sramc_test_rand_hsize HSIZE_W=2 HSIZE_R=1 N=5
31   make run TEST=sramc_test_rand_hsize DATA_CON=1 N=6
32   make run TEST=sramc_test_rand_hsize DATA_CON=2 N=7
33   make run TEST=sramc_test_rand_hsize DATA_CON=3 N=8
34   make run TEST=sramc_test_rand_hsize DATA_CON=4 N=9
35   make run TEST=sramc_test_bist                  N=10
36   make run TEST=sramc_test_bist_fail             N=11
37   make run TEST=sramc_test_rand_addr             N=12
```


49:35 / 55:45 护眼: 关 x2倍速 高清

如何验证一个32位加法器？覆盖率驱动（简单了解）

E课网 eecourse.com

随机种子

覆盖率统计

```
1  # Makefile for simulate the adder32.v
2  # Question 1. how to generate the random number?
3  # Solution 1. random seed generation
4  SEED = $(shell date +%s)
5
6  CYCLE_NUM = 200
7  # variable definition
8  DUT = ../../../design/verilog/adder32.v
9  #TB  =./adder32_tb.v ./timescale.v
10 TB  = ./adder32_tb_random.v
11 # -------- run the simualtion throught the common methods -------
12 run: compile simulate
13 compile:
14   vcs -timescale=1/1 -debug_all $(DUT) $(TB) -l com_$(SEED).log
15 simulate:
16   ./simv +plusargs_save +seed=$(SEED) +cycle_num=$(CYCLE_NUM) -l sim_$(SEED).log
17 # -------- coverage driven strategy-----------------------------
18 run_cov: compile_coverage simulate_coverage
19 compile_coverage:
20   vcs -debug_all -cm line+cond+fsm+tgl+branch -lca $(DUT) $(TB) -l com_$(SEED).log
21 simulate_coverage:
22   ./simv +plusargs_save +seed=$(SEED) +cycle_num=$(CYCLE_NUM) -cm line+cond+fsm+tgl+branch -lca -cm_log \
23     cm_$(SEED).log -l sim_$(SEED).log
24 report_cov:
25   urg -dir simv.vdb -format both -report coverage
26 dve_cov:
27   dve -cov -covdir simv.vdb -lca
28 clean:
29   @-rm -rf *.log  csrc simv simv.daidir ucli.key DVEfiles *vpd simv.vdb coverage *.bak *.help
```

```
make run_cov –f Makefile.vcs
make report_cov –f Makefile.vcs
make dve_cov –f Makefile.vcs
```

1. 先run_cov产生覆盖率数据库
2. 然后report_cov，生成txt或者html格式的覆盖率统计信息
3. 最后可以使用dve产看覆盖率

49:38 / 55:45 护眼: 关 x2倍速 高清

Makefile (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim) - GVIM1

File Edit Tools Syntax Buffers Window UVM Verilog Help

```
19    +incdir+../env \
20    ../agent/ahb/ahb_pkg.sv \
21    ../agent/ahb/ahb_if.sv \
22    ../env/ahb_sramc_uvmtb_env_pkg.sv \
23    ../tests/ahb_sramc_uvmtb_test_pkg.sv \
24    ../tb/ahb_sramc_uvmtb_tb.sv \
25    -l comp.log -kdb
26
27  run:
28    ./simv +UVM_TESTNAME=${TEST} +UVM_VERBOSITY=${VERB} +ntb_random_seed=${SEED} -l ${TEST}.log
29
30  dve:
31    dve -vpd vcdplus.vpd&
32
33  clean:
34    rm -rf csrc simv* *.log DVE* *.vpd vc_hdrs.h *.key novas* verdiLog ahb_top.fsdb
35
36  verdi:
37    verdi -ssf ahb_top.fsdb &
38
39
40  regression:
41    make clean
42    make comp
43    make run TEST=ahb_sramc_uvmtb_demo_test
44    make run TEST=ahb_sramc_uvmtb_demo_test01
45    make run TEST=ahb_sramc_full_addr_space_AA_55_data_test
46    make run TEST=ahb_sramc_full_addr_space_random_data_test
47    make run TEST=ahb_sramc_random_wr_rd_test
~
~
"Makefile" 47L, 1044C written                                                    47,42-43      Bot
```

verwcj@VM01:~/kelle... | teaching_sim | ahb_sramc_test_pkg.s... | coverage.cfg (~/keller... | Makefile (~/keller_wor... | ahb_sramc_uvmtb_tes...